Geometry & Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Rational exponents**

Consider the function $f(x) = 3x^2 - 2x + 5$. Find the value of $f'(1)$.

The derivative of a function $f(x)$ gives the instantaneous rate of change of the function at any given point. We first find the derivative of $f(x) = 3x^2 - 2x + 5$ using the power rule: $f'(x) = \frac{d}{dx}(3x^2) - \frac{d}{dx}(2x) + \frac{d}{dx}(5)$. Applying the power rule, $\frac{d}{dx}(ax^n) = nax^{n-1}$ and the derivative of a constant is 0, we get $f'(x) = 2(3x^{2-1}) - 1(2x^{1-1}) + 0$, which simplifies to $f'(x) = 6x - 2$. To find the value of the derivative at $x = 1$, we substitute $x = 1$ into the express.

2 **[Foundational]** • **Target: Exponential functions**

A circle with centre O has radius $r = 8$ cm. Chord AB subtends a central angle $\angle AOB = \frac{3\pi}{4}$ radians. (a) Find the exact length of the minor arc AB. (b) Find the exact area of the sector OAB. (c) Find the exact area of the minor segment enclosed between chord AB and arc AB.

(a) Arc length: $s = r\theta = 8 \times \frac{3\pi}{4} = 6\pi$ cm. (b) Sector area: $A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{1}{2}(8^2)(\frac{3\pi}{4}) = 32 \times \frac{3\pi}{4} = 24\pi$ cm^2. (c) Triangle area: $A_{\text{triangle}} = \frac{1}{2}r^2\sin(\frac{3\pi}{4}) = \frac{1}{2}(64)(\frac{\sqrt{2}}{2}) = 16\sqrt{2}$ cm^2. Segment area.

3 **[Foundational]** • **Target: Applications of exponential functions**

Consider the function $f(x) = e^{2x} - 4e^x + 3$. Let $u = e^x$. Find the minimum value of the function in terms of u.

Let the given function be $g(u) = u^2 - 4u + 3$, where $u = e^x$. Since $e^x > 0$, the domain for u is $u > 0$. To find the minimum value of the quadratic function $g(u)$, we can find the vertex. The u-coordinate of the vertex is given by $-\frac{b}{2a} = -\frac{-4}{2(1)} = 2$. Since $u = 2$ is within the domain $u > 0$, the minimum value occurs at $u = 2$. Substituting $u = 2$ into $g(u)$, we get $g(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the minimum value of the function in terms of u is -1.

4 **[Foundational]** • **Target: The natural exponent**

The first term of a geometric sequence is $u_1 = 18$ and the common ratio is $r = \frac{1}{3}$. (a) Find the exact value of the 5th term u_5. (b) Calculate the sum of the first 6 terms, S_6. (c) State why the sum to infinity S_∞ exists, and find its exact value.

(a) $u_5 = u_1r^4 = 18(\frac{1}{3})^4 = 18(\frac{1}{81}) = \frac{18}{81} = \frac{2}{9}$. (b) $S_6 = \frac{u_1(1-r^6)}{1-r} = \frac{18(1-(1/3)^6)}{1-1/3} = \frac{18(1-1/729)}{2/3} = 27(\frac{728}{729}) = \frac{728}{27} \approx 26.96$. (c) Since $|r| = |\frac{1}{3}| < 1$, the terms converge to 0 and the infinite series converges. $S_\infty = \frac{u_1}{1-r} = \frac{18}{1-1/3} = \frac{18}{2/3} = 27$.

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5 [Foundational] • Target: Logarithms

Consider the function $f(x) = 2x - 1$. Find the value of $f(3)$.

$f(x) = 2x - 1$. To find $f(3)$, substitute $x = 3$ into the function: $f(3) = 2(3) - 1$. $f(3) = 6 - 1$. $f(3) = 5$. Therefore, the value of $f(3)$ is 5. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

6 [Foundational] • Target: Logarithm laws and change of base

Group A has 20 data values with mean 15. Group B has 18 data values with mean 10. Find the mean of the combined data set.

Working through this statistics step: Combined mean $= \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{(20)(15) + (18)(10)}{20 + 18} = \frac{240}{19}$. Final Answer: $\frac{240}{19}$

7 [Foundational] • Target: Rational exponents

Points $A(1, 5)$ and $B(-2, 0)$ are given. Find the gradient of the perpendicular bisector of $[AB]$.

Working through this modelling step: Gradient of AB: $\frac{-5}{-3}$. Perpendicular gradient $= -\frac{1}{\text{grad}_{AB}} = -\frac{3}{5}$. Final Answer: $-\frac{3}{5}$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Exponential functions

Points $A(-2, 3)$ and $B(-3, 0)$ are given. Find the gradient of the perpendicular bisector of $[AB]$.

Working through this modelling step: Gradient of AB: $\frac{-3}{-1}$. Perpendicular gradient $= -\frac{1}{\text{grad}_{AB}} = -\frac{1}{3}$. Final Answer: $-\frac{1}{3}$

9 [Intermediate] • Target: Applications of exponential functions

Points $A(2, -1)$ and $B(7, 5)$ are given. Find the gradient of the perpendicular bisector of $[AB]$.

Working through this modelling step: Gradient of AB: $\frac{6}{5}$. Perpendicular gradient $= -\frac{1}{\text{grad}_{AB}} = -\frac{5}{6}$. Final Answer: $-\frac{5}{6}$

10 [Intermediate] • Target: The natural exponent

The area of a sector is 3 times its arc length for a circle of radius r. Find r.

Area $= 12r^2$, arc length $= r\theta$. Setting area $= 3 \times$ arc length: $12r^2 = 3r12r = 3r = 6$ (dividing both sides by $r\theta eq0$).

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11 [Intermediate] • Target: Applications of exponential functions

For the model $f(x) = 6x + 8$ in applications of exponential functions, evaluate $f(5)$.

Final Answer: 38. Substitute $x = 5$: $f(5) = 6(5) + 8 = 38$.

12 [Intermediate] • Target: Applications of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

Final Answer: $x = 4$ and $x = 9$. Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

13 [Intermediate] • Target: The natural exponent

For the model $f(x) = 2x + 9$ in the natural exponent, evaluate $f(2)$.

Final Answer: 13. Substitute $x = 2$: $f(2) = 2(2) + 9 = 13$.

14 [Intermediate] • Target: The natural exponent

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

Final Answer: $x = 1$ and $x = 3$. Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The natural exponent • Cross-Lesson: 2.01 Functions

A data set recorded for the natural exponent consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 18.5. The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

16 [Challenging] • Target: The natural exponent • Cross-Lesson: 4.01 Statistics & Probability

For the model $f(x) = 2x + 9$ in the natural exponent, evaluate $f(3)$.

Final Answer: 15. Substitute $x = 3$: $f(3) = 2(3) + 9 = 15$.

17 [Challenging] • Target: The natural exponent • Cross-Lesson: 5.01 Calculus

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

Final Answer: $x = 2$ and $x = 4$. Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

18 [Challenging] • Target: The natural exponent • Cross-Lesson: 6.01 AI SL — Specific Topics

For the model $f(x) = 2x + 9$ in the natural exponent, evaluate $f(4)$.

Final Answer: 17. Substitute $x = 4$: $f(4) = 2(4) + 9 = 17$.

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19 [Challenging] • Target: The natural exponent • Cross-Lesson: 1.01 Number & Algebra

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

Final Answer: $x = 3$ and $x = 5$. Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

20 [Challenging] • Target: The natural exponent • Cross-Lesson: 2.01 Functions

For the model $f(x) = 2x + 9$ in the natural exponent, evaluate $f(5)$.

Final Answer: 19. Substitute $x = 5$: $f(5) = 2(5) + 9 = 19$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Rational exponents

For the model $f(x) = 4x + 6$ in rational exponents, evaluate $f(2)$.

a. 14 ✓

b. 18

c. 8

d. 18

Substitute $x = 2$: $f(2) = 4(2) + 6 = 14$.

22 [Foundational] • Target: Rational exponents

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

23 [Foundational] • Target: Rational exponents

For the model $f(x) = 4x + 6$ in rational exponents, evaluate $f(5)$.

a. 26 ✓

b. 30

c. 20

d. 30

Substitute $x = 5$: $f(5) = 4(5) + 6 = 26$.

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24 [Foundational] • Target: Exponential functions

For the model $f(x) = 5x + 7$ in exponential functions, evaluate $f(2)$.

a. 17 ✓

b. 22

c. 10

d. 21

Substitute $x = 2$: $f(2) = 5(2) + 7 = 17$.

25 [Foundational] • Target: Exponential functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

26 [Foundational] • Target: Exponential functions

For the model $f(x) = 5x + 7$ in exponential functions, evaluate $f(5)$.

a. 32 ✓

b. 37

c. 25

d. 36

Substitute $x = 5$: $f(5) = 5(5) + 7 = 32$.

27 [Foundational] • Target: Applications of exponential functions

For the model $f(x) = 6x + 8$ in applications of exponential functions, evaluate $f(2)$.

a. 20 ✓

b. 26

c. 12

d. 24

Substitute $x = 2$: $f(2) = 6(2) + 8 = 20$.

28 [Foundational] • Target: Applications of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Rational exponents**

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ **and** $x = 4$ ✓

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

30 **[Intermediate]** • **Target: Rational exponents**

A data set recorded for rational exponents consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

31 **[Intermediate]** • **Target: Rational exponents**

For the model $f(x) = 4x + 6$ in rational exponents, evaluate $f(3)$.

a. 18 ✓

b. 22

c. 12

d. 22

Substitute $x = 3$: $f(3) = 4(3) + 6 = 18$.

32 **[Intermediate]** • **Target: Rational exponents**

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

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33 [Intermediate] • Target: Rational exponents

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

34 [Intermediate] • Target: Exponential functions

A data set recorded for exponential functions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

35 [Intermediate] • Target: Exponential functions

For the model $f(x) = 5x + 7$ in exponential functions, evaluate $f(3)$.

a. 22 ✓

b. 27

c. 15

d. 26

Substitute $x = 3$: $f(3) = 5(3) + 7 = 22$.

36 [Intermediate] • Target: Exponential functions

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

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37 [Intermediate] • Target: Exponential functions

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applications of exponential functions • Cross-Lesson: 6.01 AI SL — Specific Topics

A data set recorded for applications of exponential functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

39 [Challenging] • Target: Applications of exponential functions • Cross-Lesson: 1.01 Number & Algebra

For the model $f(x) = 6x + 8$ in applications of exponential functions, evaluate $f(3)$.

a. 26 ✓

b. 32

c. 18

d. 30

Substitute $x = 3$: $f(3) = 6(3) + 8 = 26$.

40 [Challenging] • Target: Applications of exponential functions • Cross-Lesson: 2.01 Functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

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41 [Challenging] • Target: The natural exponent • Cross-Lesson: 4.01 Statistics & Probability

A data set recorded for the natural exponent consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

42 [Challenging] • Target: Logarithms • Cross-Lesson: 5.01 Calculus

A data set recorded for logarithms consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

43 [Challenging] • Target: Logarithms • Cross-Lesson: 6.01 AI SL — Specific Topics

For the model $f(x) = 3x + 3$ in logarithms, evaluate $f(3)$.

a. 12 ✓

b. 15

c. 9

d. 16

Substitute $x = 3$: $f(3) = 3(3) + 3 = 12$.

44 [Challenging] • Target: Logarithm laws and change of base • Cross-Lesson: 1.01 Number & Algebra

A data set recorded for logarithm laws and change of base consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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45 [Challenging] • Target: Logarithm laws and change of base • Cross-Lesson: 2.01 Functions

For the model $f(x) = 4x + 4$ in logarithm laws and change of base, evaluate $f(3)$.

a. 16 ✓

b. 20

c. 12

d. 20

Substitute $x = 3$: $f(3) = 4(3) + 4 = 16$.

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